Power Supply LUT Method Comparison

Date: 3/27/2026

Objective

This document compares raw and cleaned power-supply LUT outputs for each cleaning method and shows how each cleaning policy changes the LUT shape.

Raw Data Source

Note on the origin of the current datasets:

- Cbmark_Beam_A_07_2025.csv was taken from the Google Sheet used during the July 11, 2025 experiment.

Google Sheet link: https://docs.google.com/spreadsheets/d/1T73CYgSkFAcI7QR085y3sOdtns9vNCIwv82ZkwA

Current raw data columns:

- beam_current
- voltage
- heater_current

Note: A temporary update was made to `data/lut/clean.py` to run and compare all three cleaning methods during this evaluation.

Input and Outputs

- Raw input file: power_supply/Raw_Cbmark_Beam_A_07_2025.csv
- max_beam output: power_supply/Raw_Cbmark_Beam_A_07_2025___max_beam.csv
- min_current_95pct_beam output: power_supply/Raw_Cbmark_Beam_A_07_2025___min_current_95pct_beam.csv
- median_top_band output: power_supply/Raw_Cbmark_Beam_A_07_2025___median_top_band.csv

Shared Pre-Processing (Applies to All 3 Methods)

All three methods use the same preparation steps before per-voltage selection:

1. Parse each CSV row to numeric values: beam_current, voltage, heater_current.
2. Drop rows with missing/non-numeric values.
3. Drop rows with any negative value.
4. Drop non-operational rows where beam_current <= 0.001 mA, voltage <= 0.001 V, or heater_current <= 0.001 A.
5. Round voltage and heater_current to 2 decimals.
6. Sort all remaining rows by (heater_current, voltage, beam_current).

7. Group all sorted rows by voltage.

Important interpretation:

- Per-voltage selection happens over the full post-filtered, sorted dataset.
- So "max at a voltage" means max in that full retained dataset after non-operational filtering.

Cathode Heating Runtime Lookup Order (with Units)

The cleaned LUT is loaded by Cathode Heating with columns:

- beam_current (mA)
- voltage (V)
- heater_current (A)

Runtime lookup flow:

1. User sets heater current (A) or heater voltage (V).
2. If current is set, code may look up voltage from current via heater_current -> voltage.
3. If voltage is set, code may look up current from voltage via voltage -> heater_current.
4. Beam prediction is obtained from voltage via voltage -> (heater_current, beam_current).
5. Grid-current prediction is derived from beam/emission model, not from a direct LUT key lookup:
   - emission_current (mA) = beam_current / 0.72
   - grid_current (mA) = 0.28 * emission_current

Key constraint for runtime safety:

- Cathode Heating expects functional mappings (unique values) for both directions used at runtime.
- In practice this means cleaned LUTs should keep one row per voltage and avoid ambiguous heater_current -> voltage mappings.

Method 1 Summary (Shown First Below)

- Method: max_beam
- Rule: For each voltage bin in the retained post-filtered dataset, keep the row with the highest beam_current.
- Why: practical performance-first approximation.
- Output behavior: one output row per voltage after filtering non-operational rows.

Method 1: max_beam

Explanation:

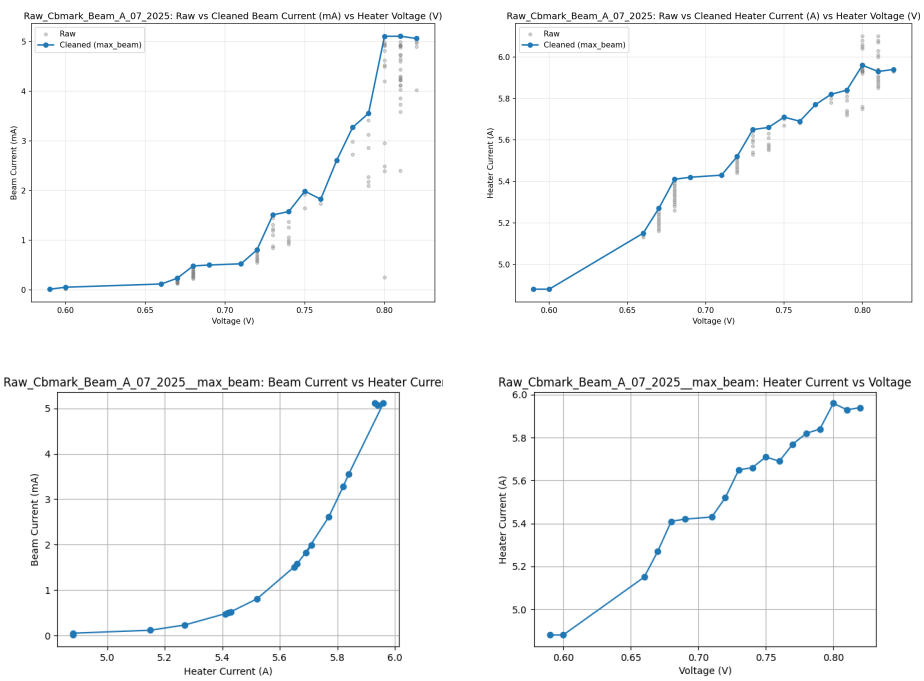
- Algorithm:

1. Run the shared pre-processing above.
2. For each voltage bin, select the row with maximum beam_current.
3. If there is a tie in beam_current, pick the one with higher heater_current.
4. Emit one cleaned row per voltage, sorted by voltage.

- Goal: practical performance-first approximation.
- Result: one output row per voltage after filtering non-operational rows.

Graphs

Raw and cleaned overlay:



Beam vs Voltage (colored by heater):

Method 2: min_current_95pct_beam

Explanation:

- Algorithm:

1. Run the shared pre-processing above.
2. For each voltage bin, compute max_beam in that bin.
3. Compute threshold = 0.95 * max_beam.
4. Keep candidate rows with beam_current >= threshold.

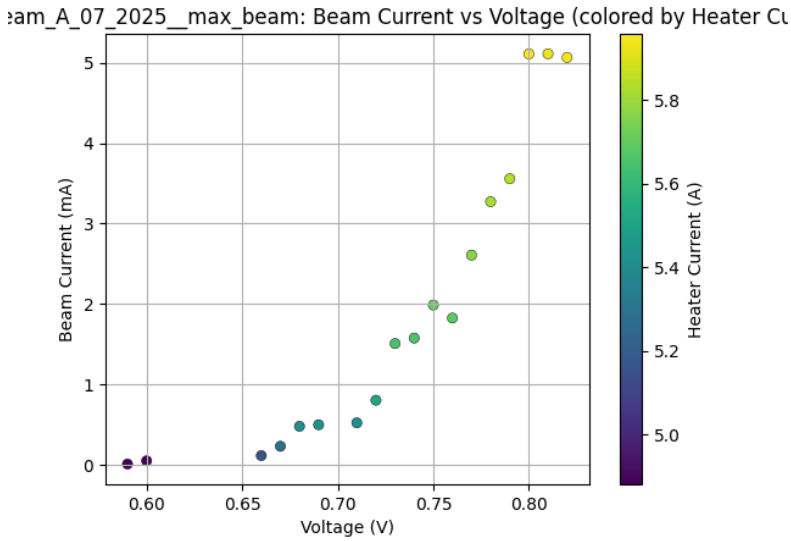


Figure 1: max_beam_clean_beam_vs_voltage_color

5. From candidates, select the row with minimum heater_current.
6. If there is a tie in heater_current, pick the one with higher beam_current.
7. Emit one cleaned row per voltage, sorted by voltage.

- Goal: near-peak beam with lower heater burden.
- Result: one output row per voltage after filtering non-operational rows.

Graphs

Raw and cleaned overlay:

Beam vs Voltage (colored by heater):

Method 3: median_top_band

Explanation:

- Algorithm:

1. Run the shared pre-processing above.
2. For each voltage bin, compute the median of beam_current values.
3. Keep the top-band rows where beam_current >= bin median.
4. Compute the median heater_current of that top band.
5. Select the top-band row with heater_current closest to the top-band median heater_current.
6. If there is a tie in distance, pick the one with higher beam_current.

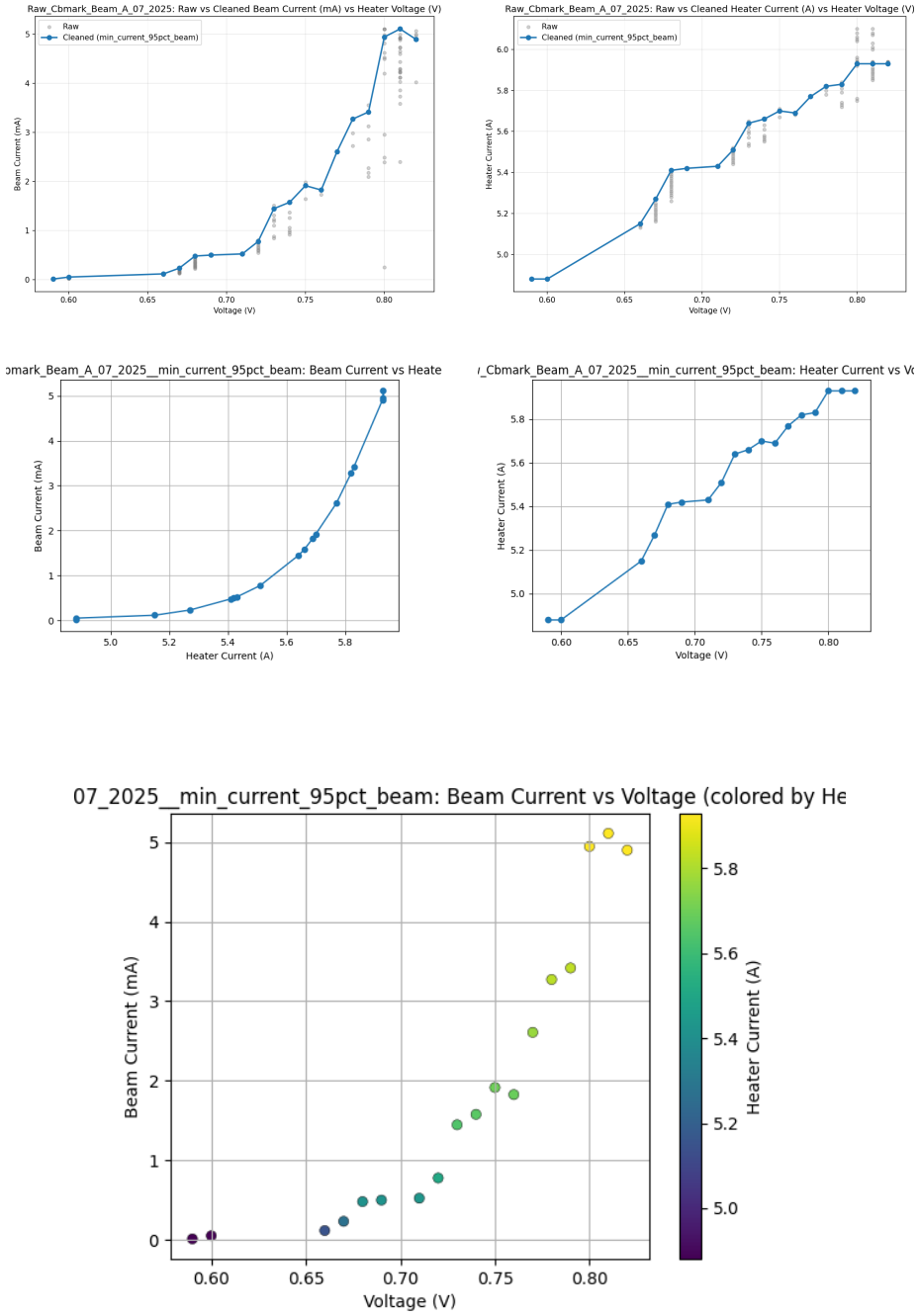


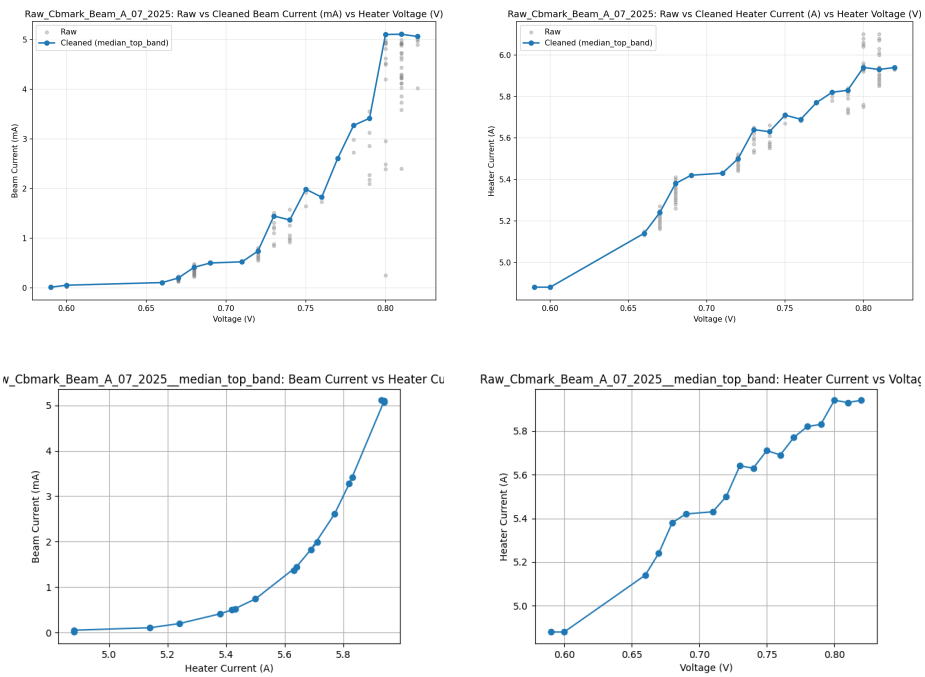
Figure 2: min__current__clean__beam__vs__voltage__color

7. Emit one cleaned row per voltage, sorted by voltage.

- Goal: robust representative point when a voltage bin has noisy clusters.
- Result: one output row per voltage after filtering non-operational rows.

Graphs

Raw and cleaned overlay:



Beam vs Voltage (colored by heater):

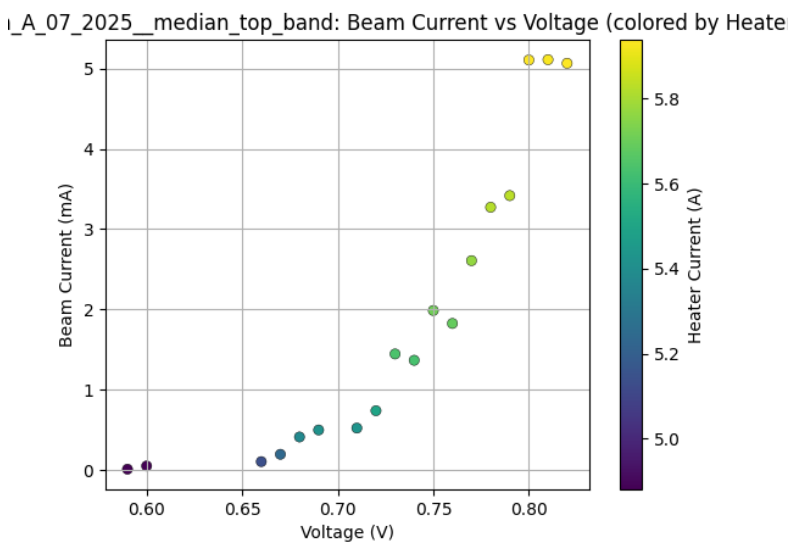


Figure 3: median_top_band_clean_beam_vs_voltage_color